

ONLINE RETAILER CUSTOMER DATA (BRAZIL-OLIST)

Using Power Bi and Orange ML To Improve On Time
Delivery

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INTRODUCTION

Brazil's Growing E-Commerce Sector

- Largest market in Latin America with rapid post-pandemic growth
- Surge in online shopping across urban and rural regions

Rising Customers Expectations

- Demand for same-day/next-day delivery
- Delays lead to poor customer satisfaction and brand loss

Importance of Delivery Optimisation

- On-time delivery is a key success factor
- Predicting and preventing delays improves efficiency

Tools used

- Power BI- for data analysis and interactive dashboards
- Orange ML - for building machine learning models to predict delivery delays



Problem Statement

Core Problem

Online retail service providers in Brazil face challenges delivering products on time.

(Problem in detail - Online retail service providers in Brazil face the challenge of delivering products to their customers on time due to various factors one being increased freight costs.)



BUSINESS USE CASE

Business problem:

Online retailers in Brazil struggle with delayed deliveries and rising freight costs, mainly due to geographic challenges and inefficient logistics.

Objective:

To identify key factors affecting delivery time and freight cost using Olist data and create visual insights for better logistics decisions

Value proposition:

Retailers gain data driven insights to reduce freight expenses, improve delivery performance, and boost customer satisfaction.



DATA OVERVIEW AND PRE PROCESSING

Rows: 1,000 observations

Columns: Includes customer_id, customer_city, customer_state, Freight_value, Seller_id, Seller_state, Price, Product_height, etc.

Missing Values: No Missing or empty dataset

Data Types: Mostly categorical (object/text fields for IDs, cities, states, etc.)

Column Standardization: Renamed columns using lowercase and underscores (e.g., customer_city) and removed special characters for consistency.

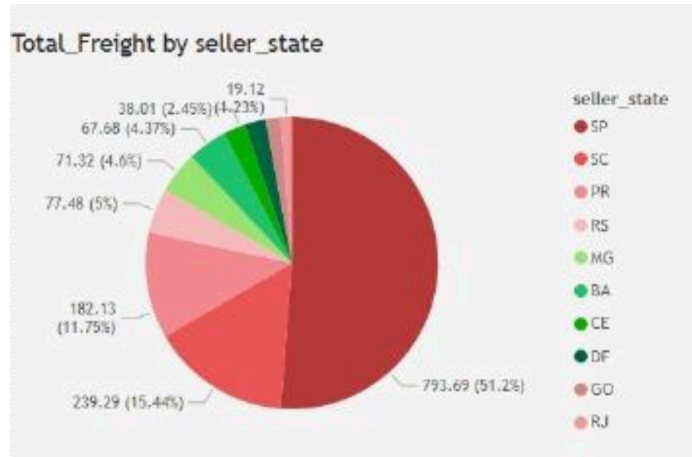
Missing Values: Checked for and resolved any null values. The final dataset had no missing data.

Data Type Correction: Ensured correct types—categorical for text fields, numeric for amounts, and converted dates to datetime format.

Filtering & Deduplication: Removed duplicate entries based on unique IDs like customer_id and filtered to a clean sample of 1,000 rows.

Table Merging: Joined related tables (e.g., orders, freight) using common keys to enrich the dataset for deeper analysis.

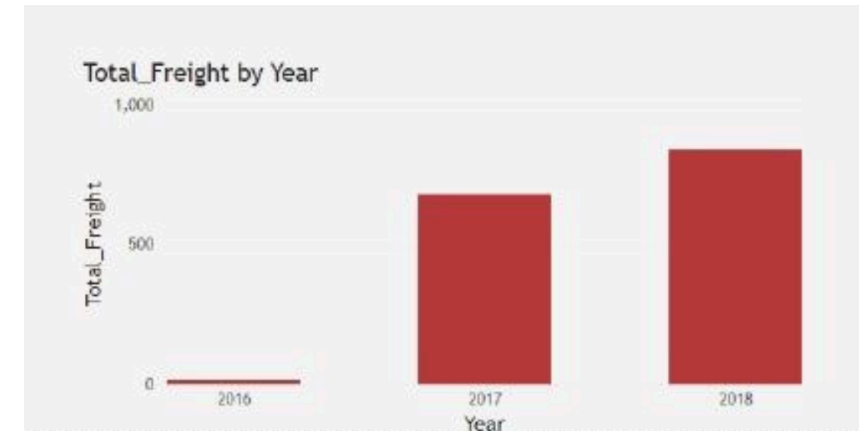
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SP accounts for 51.2% of total freight costs, likely due to centralized dispatching. Using multiple seller zones can reduce long-distance shipping and lower freight costs.



Most orders fall within 0–40 days, but outliers show high delays and freight costs, likely due to logistic inefficiencies (remote areas, warehouse delays, carriers).



Freight costs increased sharply from 2016 to 2018, indicating changes in pricing, volume, or supplier contracts over time.

Total_Freight = SUM('orders item dataset'[freight_value])
Average_Freight = AVERAGE('orders item dataset'[freight_value])

```
Delivery_Days =  
DATEDIFF (  
    'olist_orders_dataset'[order_purchase_timestamp],  
    'olist_orders_dataset'[order_delivered_customer_date],  
    DAY)
```

```
Average_Freight = AVERAGE ( 'orders item dataset'[freight_value] )
```

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customer_state	Average_Delivery_Days
AC	21.00
AL	24.50
AM	26.36
AP	27.18
BA	19.28
CE	21.20
DF	12.90
ES	15.72
GO	15.54
MA	21.51
MG	11.94
MS	15.54
MT	18.00
PA	23.73
PB	20.39
PE	18.40
PI	19.19

Delivery Days

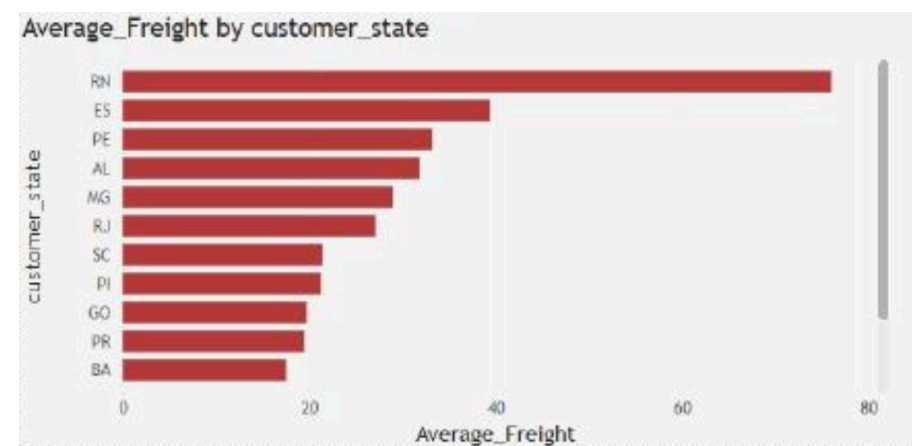
- Slow deliveries in AP, AM, AL (over 24 days).
- Fast deliveries in DF, MG (around 12 days).
- Remote areas take longer—need better local delivery setups.

Average_Delivery_Days = AVERAGE('olist_orders_dataset'[Delivery_Days])



Freight vs. Delivery Days

- Most orders take 5–25 days with average freight around 10–25 BRL.
- Some long deliveries (35–80 days) have very high freight.
- These outliers show delivery issues or costly routes.



Freight by State

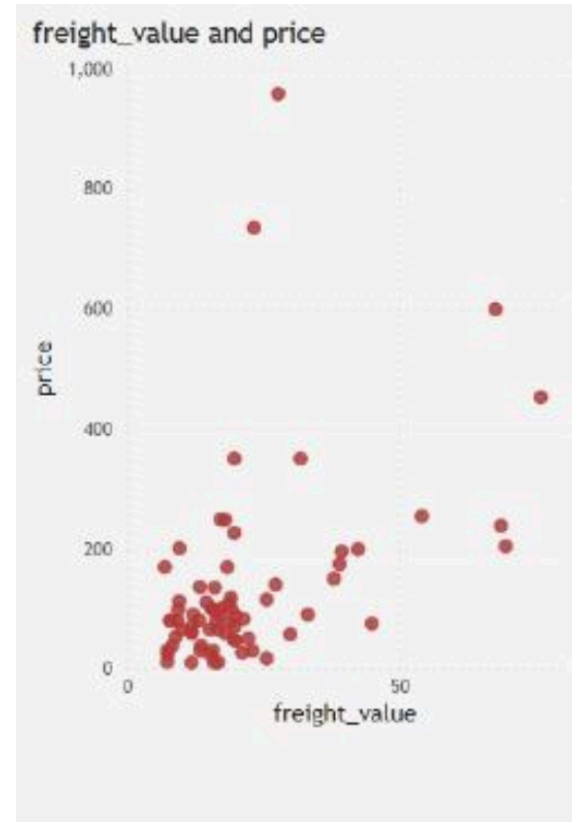
- Highest freight in RN, ES, PE (above 30 BRL).
- Lower freight in BA, PR (under 20 BRL).
- Target high-cost states for route optimization or carrier deals.

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Map Average Delivery Days & Freight Value by State

- Darker red states in Brazil mean longer delivery times and higher freight costs.
- These areas are likely far from warehouses or have fewer delivery options.
- Problem: Expensive and slow deliveries in remote regions.



Scatter Plot Freight Value vs Price

- Most products are under \$200, but some have high freight charges (over \$40).
- Problem: Shipping cost is high even for low-cost items.
- Solution: Offer free shipping for big orders or encourage item bundling to make shipping more cost-effective.

MACHINE LEARNING APPROACH

1. Size & Weight Drive Freight Costs

Lighter items (<10kg) typically incur lower freight (~BRL 50).

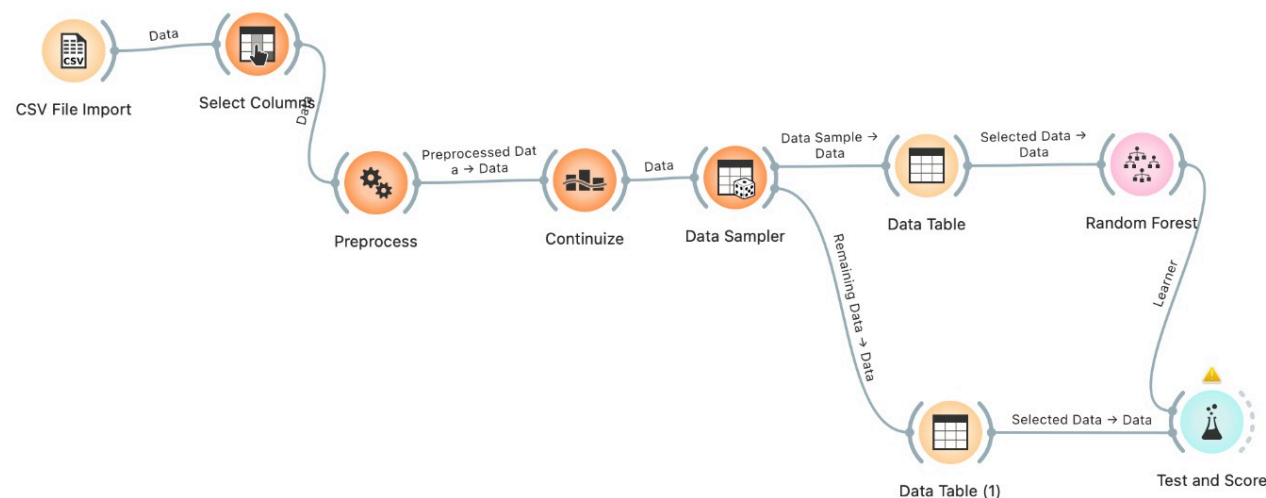
Product dimensions and packaging directly impact shipping charges.

2. Faster Delivery Costs More

There's a clear link between quicker delivery and higher freight costs.

3. Distance is a Major Factor

Interstate deliveries cost more than intrastate, showing geographic distance heavily influences freight pricing.



Key Takeaways

1. Weight, Size & Packaging Matter

Heavier and bulkier items cost more to ship.

We observed that product weight and volume directly impact freight cost especially efficient packaging could help reduce this.

2. Faster Delivery = Higher Cost

There's a clear trade-off between speed and shipping expense.

Orders with shorter delivery times often came with higher freight charges, showing that urgency drives cost.

3. Distance is the Biggest Factor

The greater the distance between the seller and customer, the higher the freight cost.

This was especially noticeable in deliveries to remote or rural regions in Brazil, where logistics get more complicated.



Challenges

- **Complex Data Integration :**

We worked with multiple interconnected dataset such as orders, items, customers, sellers, and products.

It was tricky to join them using IDs like `order_id` , `customer_id` , and `seller_id` without duplicating rows or losing valuable information.

We had to carefully ensure each join was either one-to-many or many-to-one relationship .

- **Cleaning and Formatting Issues :**

Some columns especially freight value, price, and delivery timestamps , had missing values or inconsistent formats.

For example, order dates came in mixed date time formats, which made calculations every more complex and required extra cleaning.

- **Visual Representation Challenges:**

Deciding how to best show relationships like freight cost vs. delivery time or freight vs. product price wasn't easy.

Scatter plots seemed ideal, but they quickly got cluttered with overlapping data points and outliers, making insights hard to spot.



REFERENCES

- NUS Online Retail Customer Dataset
- Power Bi(Power query, Data Cleaning, Dashboard)
- Machine Learning (Orange), Data Mining
- Microsoft Excel





THANK YOU